MemoGuard — AI-Powered Alzheimer's & Dementia Care Platform

Comprehensive Mobile App Development Prompt

🧠 Project Overview

Build **MemoGuard**, a dual-sided AI-powered mobile care platform consisting of:

1. **Patient App** — Simplified, voice-first, large-element mobile experience for Alzheimer's/dementia patients
2. **Caregiver App / Web Dashboard** — Comprehensive monitoring, scheduling, and alert management interface for family caregivers and clinical staff

The platform uses Claude AI (Anthropic) at its core for intelligent reminders, behavioral analysis, cognitive game scoring, and clinical insights.

🎨 Design System

Patient App — Visual Language

- **Color Palette**: Deep indigo `#3B2F8F` (primary), calm sky blue `#5BAFD6` (accent), soft lavender `#C3B8E8` (background tints), pure white `#FFFFFF` (text on dark), gentle coral `#F28C7D` (emergency/alert)
- **Typography**: Display — `Nunito` (rounded, warm, highly legible); Body — `Atkinson Hyperlegible` (purpose-built for low vision). MINIMUM body text: `22sp`. Heading text: `32–48sp`.
- **Buttons**: Minimum 80px height, full-width or near-full-width, rounded corners `border-radius: 24px`, bold labels only, never icon-only
- **Spacing**: Generous — minimum 24px between interactive elements, 40px section padding
- **Motion**: Slow, gentle transitions (400–600ms ease-in-out). No sudden flashes. Subtle pulse animations on reminder alerts.
- **Accessibility**: WCAG AAA contrast ratios, voice-over compatible, touch target minimum 56×56px

Caregiver App — Visual Language

- **Color Palette**: Deep navy `#1A2340` (sidebar/header), clean white `#FFFFFF` (content), accent blue `#4A90D9`, status colors: green `#27AE60` (active), amber `#F39C12` (warning), red `#E74C3C` (emergency)

- **Typography**: `DM Sans` (interface) + `DM Mono` (data/timestamps). Clear information hierarchy.
- **Layout**: Card-based, data-dense but scannable. Fixed left sidebar on tablet/web. Bottom nav on mobile.
- **Motion**: Snappy transitions (200ms). Alert cards animate in from top. Real-time feed uses smooth insert animations.

📱 PATIENT APP — Full Screen Specifications

Tech Stack

- **Framework**: React Native (Expo) for iOS + Android
- **State**: Zustand + React Query
- **AI Layer**: Anthropic Claude API (`claude-sonnet-4-20250514`) for voice responses, daily recommendations, and behavioral analysis
- **Voice**: Expo Speech (TTS) + `@react-native-voice/voice` (STT)
- **Navigation**: React Navigation v6 (Bottom Tab — max 4 tabs)
- **Notifications**: Expo Notifications (local + push)
- **Backend**: Supabase (auth, realtime DB, storage) or Firebase
- **Maps/GPS**: React Native Maps + Expo Location

SCREEN 1: Lock / Home Screen (Primary Reminder Screen)

Purpose: The first thing the patient sees every time they pick up their phone. Designed to be impossible to miss.

UI Layout (top → bottom):

```
┌──────────────────────────────────┐
│  [Time: 08:00 AM]  [Date: Mon, Mar] │  ← Large, top-center, 32sp
│                                  │
│  ┌──────────────────────────┐    │
│  │  💊  Time for your        │    │  ← Pulsing reminder card
│  │      MORNING MEDICINE     │    │     Background: indigo gradient
│  │      Donepezil 10mg       │    │     Text: white, 36sp bold
│  └──────────────────────────┘    │
│                                  │
│  🔊 [Reads aloud automatically]  │  ← Auto-plays TTS on open
│                                  │
│  ┌──────────────────────────┐    │
│  │  ✅  TAKEN                 │    │  ← Green button, 88px height
│  └──────────────────────────┘    │
│                                  │
│  ┌──────────────────────────┐    │
│  │  ⏰  Remind me in 10 min   │    │  ← Blue button, 88px height
│  └──────────────────────────┘    │
│                                  │
│  ┌──────────────────────────┐    │
│  │  🆘  I NEED HELP           │    │  ← Coral/red button, 88px height
│  └──────────────────────────┘    │
└──────────────────────────────────┘
```

Behaviors:

- On screen wake: auto-play TTS of the current pending reminder (e.g., *"Good morning Margaret. It's time to take your Donepezil. Tap the green button when you've taken it."*)
- "Taken" → logs confirmation with timestamp to DB → notifies caregiver dashboard → shows celebratory animation + positive voice feedback ("Well done! That's your morning medicine done.")
- "Remind me in 10 min" → schedules local notification → dismisses current prompt
- "I Need Help" → opens Emergency Screen (Screen 4)
- If no response within 5 minutes → auto-escalate: push alert to caregiver

AI Integration:

```javascript
```

```
// Generate personalized reminder message
const response = await fetch("https://api.anthropic.com/v1/messages", {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: `You are a warm, patient, and gentle care assistant for an Alzheimer's patient.
    Always use simple words. Speak in short sentences. Address them by first name.
    Be encouraging and positive. Never show frustration.`,
    messages: [{
      role: "user",
      content: `Generate a gentle spoken reminder for ${patientName} to take ${medicineName}
      Current time: ${currentTime}. Keep it under 3 sentences. Be warm and reassuring.`
    }]
  })
});
```

SCREEN 2: Daily Routine Screen

Purpose: Shows the patient their full day's schedule in chronological order with visual status.

UI Layout:

- Header: "Your Day" in 36sp + today's date
- Vertical scrollable list of task cards
- Each card: 100px height, left colored bar (by category), large icon (48px), task name (28sp), time (20sp), status badge

Task Categories & Colors:

Category	Color	Icon
Medicine	Indigo `#3B2F8F`	💊
Prayer / Spiritual	Lavender `#9B7FC7`	🙏
Appointment	Blue `#5BAFD6`	📅
Exercise	Green `#4CAF82`	🏃
Memory Game	Amber `#F5A623`	🧩
Music	Pink `#E91E8C`	🎵
Hydration	Cyan `#00BCD4`	💧
Meal	Orange `#FF7043`	🍽️

Voice Feature: Each task card has a speaker icon. Tap → TTS reads the task details aloud. Auto-announce upcoming tasks 10 minutes before.

AI Daily Recommendations (generate at 6 AM daily):

```javascript
// AI generates personalized daily recommendations
const dailyPlan = await fetch("https://api.anthropic.com/v1/messages", {
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: "You are a dementia care coordinator. Generate simple, clear daily recommendatic
    messages: [{
      role: "user",
      content: `Patient: ${patientName}, Age: ${age}, Stage: ${dementiaStage}
      Today's weather: ${weather}. Recent mood: ${recentMood}.
      Yesterday's completion rate: ${completionRate}%.
      Generate 3 gentle daily tips and suggest the best time for their memory game today.
      Return as JSON: { tips: [], gameRecommendedTime: "", morningMessage: "" }`
    }]
  })
});
```

Task Completion Flow:

1. Tap task → expand with full instructions (step-by-step for complex tasks like morning routine)
2. "Mark Complete" large green button

3. Confirmation sound + checkmark animation
4. Real-time sync to caregiver dashboard

SCREEN 3: Mental Games Hub

Purpose: Cognitively stimulating games calibrated to patient's level.

Layout: 3 large game cards on a scrollable screen, each with:

- Game illustration (80×80px icon)
- Game name (28sp)
- Difficulty level (stars)
- "Last score" display
- "PLAY" button (indigo, full-width)

Game 1: Memory Match

- Grid of 4×4 cards (start with 3×3 for beginners)
- Flip cards to find matching pairs (fruits, animals, everyday objects)
- Timer displayed at top
- Score = pairs found × time bonus
- Voice: *"Find the matching pairs. Tap a card to flip it."*
- Adaptive: if >80% success 3 sessions in a row → increase grid; if <40% → reduce

Game 2: Word Recall

- Show 5 simple words for 10 seconds
- Hide words
- Ask patient to tap the words they remember from a list of 10
- Score = correct recalls / total × 100
- Voice reads each word aloud during display phase

Game 3: Simple Reaction

- Large colored shape appears on screen (3 colors: red circle, blue square, green triangle)
- Voice says *"Tap the RED one!"*
- Measure reaction time in ms
- Tracks reaction speed trends over weeks

Post-Game Flow:

```
┌──────────────────────────────┐
│  🎉  Great job, Margaret!    │
│                              │
│  Your score: 7/10            │
│  Time: 2 min 34 sec          │
│  Best score: 8/10            │
│                              │
│  [Send results to Dr. Patel] │  ← Optional, one-tap
│  [Play again]                │
│  [Go Home]                   │
└──────────────────────────────┘
```

AI Score Analysis (sent to caregiver dashboard):

```javascript
const analysis = await fetch("https://api.anthropic.com/v1/messages", {
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: "You are a clinical neuropsychologist assistant. Analyze cognitive game data.",
    messages: [{
      role: "user",
      content: `Patient game data this week:
      Memory Match scores: ${matchScores}
      Word Recall scores: ${recallScores}
      Reaction times (ms): ${reactionTimes}
      Previous month average: ${monthlyAvg}

      Provide: 1) Brief clinical interpretation, 2) Trend direction, 3) Recommended action.
      Return JSON: { interpretation, trend, recommendation, flagForDoctor: boolean }`
    }]
  })
});
```

SCREEN 4: Emergency Screen

Purpose: One-tap emergency alert. Must be reachable from every screen via persistent floating button.

Trigger: Red circular floating button (64px diameter) bottom-right corner, labeled "SOS" in bold white. Semi-transparent when idle, fully visible on tap.

Emergency Screen Layout:

```
┌──────────────────────────────────┐
│  🚨  SENDING ALERT...            │  ← Red full-screen takeover
│                                  │
│  Your location is being shared   │
│  with your family                │  ← 28sp white text
│                                  │
│  [────────────────────]          │  ← Progress bar (3 sec countdown)
│                                  │
│  📞  CALLING DAVID (Son)         │  ← Auto-initiates call
│                                  │
│  ┌──────────────────────────┐    │
│  │  Cancel (if accidental)  │    │  ← Small, gray, 3-second window
│  └──────────────────────────┘    │
└──────────────────────────────────┘
```

Fall Detection (via accelerometer):

```javascript
import { Accelerometer } from 'expo-sensors';

// Detect sudden acceleration spike indicating fall
Accelerometer.addListener(({ x, y, z }) => {
  const magnitude = Math.sqrt(x*x + y*y + z*z);
  if (magnitude > FALL_THRESHOLD) { // calibrate to ~2.5g
    triggerFallAlert({
      type: 'FALL_DETECTED',
      location: await getCurrentLocation(),
      timestamp: new Date().toISOString(),
      patientId: patient.id
    });
  }
});
```

Alert Payload sent to all caregivers:

```json
```

```
{
  "type": "EMERGENCY_SOS",
  "patientName": "Margaret Chen",
  "timestamp": "2026-03-28T14:32:00Z",
  "location": { "lat": 51.5074, "lng": -0.1278, "address": "23 Oak Street, London" },
  "trigger": "manual_sos | fall_detected | geofence_breach",
  "vitals": { "lastBP": "135/82", "heartRate": 78 },
  "message": "Margaret activated emergency alert at 2:32 PM"
}
```

SCREEN 5: Music & Calm

Purpose: Therapeutic music player with calming content, proven to reduce agitation in dementia patients.

Layout:

- Large album art (200×200px, rounded square)
- Track name in 28sp, artist in 20sp muted
- Category buttons: Calm Classical | Nature Sounds | Spiritual | 70s/80s Memories | Guided Breathing
- Large play/pause button (80px), skip buttons (64px)
- Volume slider with haptic feedback at each stop

Features:

- Playlist auto-generated based on patient's music preferences (set by caregiver)
- "Memory Music" feature: caregiver uploads songs from patient's youth/meaningful period
- Timer: auto-stop after 30/60 minutes (configurable)
- Voice: *"Playing your favourite relaxation music, Margaret."*

SCREEN 6: Settings (Patient-Controlled)

Minimal — only 2 settings the patient can change:

1. **Voice Type**: Large toggle — Female Voice / Male Voice (with preview button to hear a sample)
2. **Text Size**: Three options displayed as actual large text samples — Normal / Large / Extra Large

All other settings (medications, routines, geofence, emergency contacts) are managed exclusively through the Caregiver Dashboard.

Notification System (Patient App)

Notification Types & Rules:

Type	Trigger	Sound	Vibration	Auto-read?
Medicine reminder	Scheduled time	Gentle chime	3 pulses	YES
Hydration alert	Every 2 hours	Water droplet sound	1 pulse	YES
Prayer time	Scheduled (by caregiver)	Soft bell	2 pulses	YES
Exercise prompt	Scheduled	Upbeat tone	2 pulses	YES
Game session	AI-recommended time	Playful tone	2 pulses	YES
Appointment	1 hour + 10 min before	Calendar chime	3 pulses	YES
Fall alert received	Immediate	None	Haptic	NO
Caregiver message	Real-time	Gentle ping	1 pulse	YES

AI-Powered Smart Timing:

```javascript
// Learn patient's optimal reminder timing
const optimalTimes = await fetch("https://api.anthropic.com/v1/messages", {
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: "Analyze patient response patterns to optimize reminder timing.",
    messages: [{
      role: "user",
      content: `Patient response history for last 30 days:
      ${JSON.stringify(responseHistory)}

      Identify: 1) Times with highest task completion, 2) Times patient is most responsive,
      3) Times to avoid (confusion patterns). Return optimized schedule as JSON.`
    }]
  })
});
```

💻 CAREGIVER APP / WEB DASHBOARD — Full Screen Specifications

Tech Stack

- **Web:** Next.js 14 (App Router) + TypeScript
- **Mobile:** React Native (Expo) — companion app for on-the-go alerts
- **Styling:** Tailwind CSS + shadcn/ui
- **Charts:** Recharts + Nivo
- **Real-time:** Supabase Realtime or Socket.io
- **Maps:** Mapbox GL JS
- **AI:** Anthropic Claude API (analysis, labeling assistance, report generation)
- **PDF Reports:** jsPDF + html2canvas

SCREEN 1: Main Dashboard

Layout (3-column on desktop, 1-column collapsed on mobile):

```
┌──────────────────────────────────────────────────────────────────┐
│ [MemoGuard Logo]  Patients ▼   Schedule   Reports   Settings     │ ← Top nav
├──────────┬──────────────────────────────────────┬────────────────┤
│          │                                      │                │
│ SIDEBAR  │    PATIENT OVERVIEW CARDS            │  LIVE FEED     │
│          │                                      │                │
│ 👤 Mary  │  ┌──────────────────────────────┐    │ 08:03 ✅ Med   │
│ 🟢 Active│  │ [Photo]  Margaret Chen, 74   │    │ taken          │
│          │  │  🟢 ACTIVE — Last seen 2m ago │    │                │
│ 👤 John  │  │                              │    │ 08:15 💊 Next  │
│ 🔴 Alert │  │ Today: 6/10 tasks complete   │    │ med in 45min   │
│          │  │ Medications: 2/3 taken       │    │                │
│ + Add    │  │ Mood: 😊 Good (logged 9am)   │    │ 07:45 🎮 Game  │
│ Patient  │  └──────────────────────────────┘    │ score: 7/10    │
│          │                                      │                │
│          │  ┌──────────────────────────────┐    │ 07:00 🌼       │
│          │  │ VITALS SNAPSHOT              │    │ Routine done   │
│          │  │ BP: 128/78  ❤️ 72bpm         │    │                │
│          │  │ Temp: 36.8°C  SpO2: 97%      │    │ [Load more]    │
│          │  └──────────────────────────────┘    │                │
│          │                                      │                │
│          │  [⚠️ ALERTS]  John left safe zone    │                │
│          │  [📞 Call John]  [📍 View Map]       │                │
└──────────┴──────────────────────────────────────┴────────────────┘
```

Real-time Alert Banner (appears at top on emergency):

```
┌──────────────────────────────────────────────────────────────┐
│ 🚨  EMERGENCY: Margaret activated SOS at 2:32 PM              │
│ 📍  23 Oak Street, London  |  [📞 Call Now]  [📍 Open Map]     │
└──────────────────────────────────────────────────────────────┘
```

Patient Status Cards: Each shows —

- Profile photo (80×80px circular)
- Full name + age
- Status pill: 🟢 Active | 🟡 Idle (30+ min) | 🔴 Emergency | ⚫ Offline
- Task completion progress bar (today)
- Last check-in time
- Quick action icons: 📞 📍 💬

SCREEN 2: Reminders Scheduler

Layout: Split view — calendar/timeline on left, add/edit form on right

Add Reminder Form:

```
┌──────────────────────────────────────────────────────────┐
│  Add New Reminder                              [✕ Close]  │
├──────────────────────────────────────────────────────────┤
│  Type:    [💊 Medicine ▼]                                 │
│                                                           │
│  Name:    [Donepezil_________________]                    │
│  Dosage:  [10mg____________________]                      │
│                                                           │
│  Image:   [📷 Upload photo of pill] (shows on patient     │
│            screen for recognition)                        │
│                                                           │
│  Time:    [08:00 AM]    [+ Add another time]              │
│                                                           │
│  Repeat:  [Daily ▼]  Mon ✓ Tue ✓ Wed ✓ Thu ✓ Fri ✓        │
│                                                           │
│  Custom voice message:                                    │
│  "Time to take your [auto-filled]" ← default             │
│  [Record custom message 🎤] OR [Type custom text]         │
│                                                           │
│  Auto-confirm if no response after: [5 min ▼]            │
│  ☑ Notify me when patient confirms                        │
│  ☑ Notify me if patient does NOT confirm within 15 min    │
│                                                           │
│  [Preview on patient device]    [Save Reminder]          │
└──────────────────────────────────────────────────────────┘
```

Timeline View: Visual day grid showing all scheduled reminders, color-coded by category. Drag-and-drop to reschedule.

SCREEN 3: Location & Geofence

Map View (Mapbox):

- Patient's real-time location (animated pulsing blue dot)
- Safe zones shown as colored overlays (green = safe, yellow = caution zone, red = boundary)
- Location history trail for past 24 hours (dotted blue line)
- Last 5 location updates in sidebar with timestamp

Add Safe Zone Panel:

```
┌──────────────────────────────────────┐
│  Define Safe Zone                    │
│                                      │
│  Zone name: [Home_______________]    │
│  Type: ● Circle  ○ Custom polygon    │
│  Radius: [──●────────────] 200m      │
│                                      │
│  Alert levels:                       │
│  Early warning (150m from boundary):  │
│    ☑ Notify caregiver app            │
│  Urgent alert (boundary crossed):    │
│    ☑ Notify ALL caregivers           │
│    ☑ Auto-call patient               │
│    ☑ Sound alert on patient device   │
│                                      │
│  [Draw on map]         [Save Zone]   │
└──────────────────────────────────────┘
```

Geofence Alert Flow:

1. Patient crosses boundary → instant push to all caregivers
2. Alert card shows: patient photo, map thumbnail, time, "Left [Zone Name]"
3. Actions: 📞 Call Patient | 📱 Send Voice Message | 🚗 Get Directions | 🚨 Contact Authorities

---

SCREEN 4: Reports & Mental Scores

Layout: Tabs — Weekly | Monthly | Custom Range

Cognitive Performance Dashboard:

```
┌──────────────────────────────────────────────────────────────┐
│  Mental Health Report — Margaret Chen          [📄 PDF] [📋] │
│  Week of March 24 – 28, 2026                                 │
├──────────────────────────────────────────────────────────────┤
│  COGNITIVE TREND                                             │
│                                                              │
│  Score                                                       │
│  10 ┤                        ●                              │
│   8 ┤      ●            ●────                               │
│   6 ┤ ●────   ●─●                                           │
│   4 ┤                                                        │
│     └──┴──┴──┴──┴──┴──┴──── Days                            │
│       Mon Tue Wed Thu Fri Sat Sun                            │
│                                                              │
│  📊 Memory Match avg: 7.2/10  (+0.8 vs last week) ▲          │
│  📊 Word Recall avg:  6.5/10  (-0.5 vs last week) ▼          │
│  📊 Reaction time:    1.2s avg  (baseline: 1.4s)  ▲          │
│                                                              │
│  ⚠️  AI Flag: Word recall declined 2 weeks consecutively.    │
│     Recommend clinical review before next appointment.       │
├──────────────────────────────────────────────────────────────┤
│  MEDICATION ADHERENCE                                        │
│  ▓▓▓
```

```
const report = await fetch("https://api.anthropic.com/v1/messages", {
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: `You are a clinical AI assistant for dementia care. Generate professional,
    evidence-based clinical summaries from patient monitoring data. Use clear,
    medically appropriate language. Flag concerns appropriately.`,
    messages: [{
      role: "user",
      content: `Generate weekly clinical summary for ${patientName}:
      Game scores: ${gameData}
      Medication adherence: ${medicationData}
      Vitals readings: ${vitalsData}
      Mood entries: ${moodData}
      Incidents: ${incidents}

      Output JSON: { executiveSummary, cognitiveAssessment, medicationReview,
      moodAnalysis, recommendations, clinicalFlags, riskLevel }`
    }]
  })
});
```

PDF Report Structure:

- Cover: Patient name, photo, date range, generated by MemoGuard
- Section 1: Executive Summary (AI-generated)
- Section 2: Cognitive Performance Charts
- Section 3: Medication Adherence Log
- Section 4: Vital Signs Trends
- Section 5: Incident Log
- Section 6: Clinical Recommendations

SCREEN 5: Live Alerts & Actions

Alert Card Design:

```
┌──────────────────────────────────────────────────────────┐
│ 🚨 HIGH PRIORITY                              2:32 PM today  │
│                                                              │
│ [Photo]  Margaret activated SOS                              │
│          📍 23 Oak Street, London — Map thumbnail            │
│                                                              │
│  [📞 Call Margaret]   [🔊 Send Voice Message]                │
│  [📍 Full Map View]   [🚨 Notify Local Responders]           │
│                                          [✓ Mark Resolved]  │
└──────────────────────────────────────────────────────────┘
```

Alert Priority Levels:

Level	Color	Sound	Auto-escalate
P1: Emergency (SOS, Fall)	Red + flash	Loud siren	After 60s if no ack
P2: Geofence breach	Orange	Urgent tone	After 3 min
P3: Missed medication	Yellow	Chime	After 15 min
P4: Routine incomplete	Blue	Soft ping	No
P5: Info (task done)	Green	None	No

Voice Message Flow:

1. Caregiver taps "Send Voice Message"
2. Options: 🎙 Record live | 📝 Type (AI converts to speech in caregiver's saved voice profile)
3. Delivered to patient device immediately with auto-play

SCREEN 6: Labeling Interface (AI Training)

Purpose: Caregivers tag behavioral events to build ML training data for predictive alerts.

Layout: Chronological event feed with inline tagging:

```
┌──────────────────────────────────────────────────────────────┐
│  Event Labeling — Help train Margaret's AI model             │
├──────────────────────────────────────────────────────────────┤
│  08:34 AM — Patient did not respond to medication reminder    |
│  [+ Add Label ▼]                                             │
│   → Disorientation | Agitation | Sleeping | Refused |        │
│      Side Effect | Forgot | Confusion | Other...             │
│                                                              │
│  Notes: [Add caregiver note_________________________]        │
│                                              [Save Label]    |
├──────────────────────────────────────────────────────────────┤
│  09:15 AM — Unusual movement detected (accelerometer)        │
│  ✓ Labeled: "Restlessness — normal for this time of day"     |
│  [Edit label]                                                │
└──────────────────────────────────────────────────────────────┘
```

AI Labeling Assistant:

```javascript
// AI suggests labels based on context
const suggestion = await fetch("https://api.anthropic.com/v1/messages", {
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    system: "You are a dementia care behavioral analyst. Suggest labels for patient events."
    messages: [{
      role: "user",
      content: `Event: "${eventDescription}"
      Time: ${eventTime}, Patient history context: ${recentHistory}
      Suggest the most likely label from: [Disorientation, Agitation, Sundowning,
      Refusal, Fatigue, Normal variation, Side effect, Anxiety, Other]
      Return JSON: { suggestedLabel, confidence, reasoning }`
    }]
  })
});
```

🔗 Data Architecture

Supabase Schema (key tables)

```sql
```

```
-- Core entities
patients (id, name, photo_url, dob, diagnosis_stage, primary_caregiver_id, settings_json)
caregivers (id, name, role: 'family'|'professional'|'clinician', permissions_json)
patient_caregivers (patient_id, caregiver_id, relationship)

-- Events
medication_logs (id, patient_id, med_id, scheduled_time, confirmed_at, status, notes)
task_completions (id, patient_id, task_id, completed_at, duration_seconds)
location_events (id, patient_id, lat, lng, accuracy, zone_status, timestamp)
vital_readings (id, patient_id, bp_systolic, bp_diastolic, heart_rate, spo2, temp, recorded_
mood_entries (id, patient_id, mood_score, mood_emoji, notes, recorded_at)

-- Games
game_sessions (id, patient_id, game_type, score, max_score, duration_seconds, played_at)
cognitive_assessments (id, patient_id, assessment_date, overall_score, ai_analysis_json)

-- Alerts
alerts (id, patient_id, type, priority, payload_json, triggered_at, acknowledged_at, resolve
alert_actions (id, alert_id, caregiver_id, action_type, performed_at)

-- AI
ai_recommendations (id, patient_id, generated_at, category, content, acted_upon)
behavioral_labels (id, patient_id, event_id, label, caregiver_id, notes, created_at)
```

Real-time Subscriptions

```javascript
// Caregiver dashboard listens for patient events
supabase
  .channel(`patient-${patientId}`)
  .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'alerts' }, handleNewA
  .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'location_events' }, h
  .on('postgres_changes', { event: 'UPDATE', schema: 'public', table: 'medication_logs' }, h
  .subscribe();
```

🔒 Security & Compliance

- **Encryption:** AES-256 at rest, TLS 1.3 in transit
- **Auth:** Supabase Auth with MFA for caregivers/clinicians
- **Row Level Security:** Patients only see own data; caregivers see assigned patients only
- **Audit Log:** Every data access logged with user ID + timestamp

- **HIPAA Compliance**: BAA with Supabase, no PHI in logs, data residency controls
- **Patient Privacy**: Patients control which data is shared with each caregiver via granular toggles (managed by caregiver app during setup)
- **Data Retention**: Configurable — default 2 years, exportable at any time

---

📋 Development Phases

Phase 1 — Core (Weeks 1-6)

- [ ] Auth system (all 3 roles)
- [ ] Patient: Lock/Home screen + medication reminders
- [ ] Patient: Daily routine screen
- [ ] Caregiver: Dashboard + live feed
- [ ] Caregiver: Reminders scheduler
- [ ] Push notifications (local + push)
- [ ] Basic AI integration (reminder messages)

Phase 2 — Safety (Weeks 7-10)

- [ ] Emergency SOS button + fall detection
- [ ] GPS tracking + geofencing
- [ ] Real-time caregiver alerts
- [ ] Location history map view

Phase 3 — Cognitive (Weeks 11-14)

- [ ] Memory Match game
- [ ] Word Recall game
- [ ] Reaction game
- [ ] Game scoring + AI analysis
- [ ] Score trends + caregiver dashboard integration

Phase 4 — Intelligence (Weeks 15-18)

- [ ] AI daily recommendations
- [ ] AI report generation + PDF export
- [ ] Behavioral labeling interface
- [ ] Cognitive decline prediction model
- [ ] Voice assistant (STT/TTS integration)

Phase 5 — Polish (Weeks 19-20)

- [ ] Music player + therapeutic content
- [ ] Accessibility audit (WCAG AAA)

- [ ] Performance optimization
- [ ] Beta testing with care homes
- [ ] App Store + Play Store submission

💡 Key AI Prompt Patterns

1. Personalized Reminder Voice

```
System: "You are a warm care assistant for an Alzheimer's patient. Always use their first name. Keep sentences short. Be encouraging."
User: "Generate a 2-sentence spoken reminder for [name] to [task] at [time]. Make it feel like a kind friend, not a robot."
```

2. Cognitive Trend Analysis

```
System: "You are a neuropsychology AI. Analyze cognitive game data for clinical relevance."
User: "Compare this week's scores vs 4-week baseline. Flag any concerning trends. Output: {trend, severity, recommendation, clinicalFlag}"
```

3. Daily Care Recommendations

```
System: "Generate personalized dementia care recommendations based on real patient data."
User: "Patient profile + last 7 days activity + mood + weather → recommend: best game time, hydration tips, activity suggestion, caregiver note"
```

4. Incident Label Suggestion

```
System: "You are a dementia behavioral analyst. Classify patient behavioral events."
User: "Event context + patient history → suggest behavioral label + confidence score + brief reasoning"
```

5. Clinical Report Generation

```
System: "Generate professional clinical summaries from dementia patient monitoring data."
User: "All patient data for date range → produce structured clinical summary with executive summary, findings, red flags, recommendations"
```

Built for the 55 million people living with dementia worldwide and the families who love them.
MemoGuard — Because every day matters.